

The problem requires counting monic degree 7 polynomials with integer coefficients satisfying three conditions: all coefficients have odd absolute values less than 2025^2 , and every absolute value of a coefficient is a root of the polynomial.

First, denote the polynomial as $P(x) = x^7 + a_6x^6 + \cdots + a_0$, where each a_i is an integer. The absolute values of all coefficients, including the leading coefficient 1, must be roots of P . By the integer root theorem, each root (an absolute value of a coefficient) must divide the constant term a_0 .

Considering the constraints, the set S of absolute values of coefficients must satisfy:

1. $|S| \leq 7$ since P has degree 7.
2. All elements of S are odd positive integers less than 2025^2 .
3. The least common multiple of S equals $|a_0|$.

If S contains any element greater than 1, constructing such a polynomial is highly restrictive. For example, introducing a coefficient like 3 would require $P(3) = 0$, but expanding polynomials with roots including 3 typically produces coefficients with absolute values not in S . This leads to contradictions, suggesting no solutions exist for $|S| > 1$.

Thus, the only feasible case is $S = \{1\}$, where all coefficients are ± 1 . For $P(1) = 0$, the sum of coefficients must be zero. The leading term is $+1$, so the sum of a_6 to a_0 must be -1 . With seven coefficients, choosing three $+1$ s and four -1 s achieves this sum. The number of ways to choose three positions out of seven for $+1$ s is $\binom{7}{3} = 35$.

Each such polynomial meets all conditions: coefficients are ± 1 (odd, less than 2025^2), their absolute values (1) are roots since $P(1) = 0$, and no other roots or coefficient violations exist.

Thus, the number of good polynomials is 35.